HR WORKFORCE ANALYTICS PROJECT REPORT

Employee Attrition, Satisfaction, Compensation, and Career Growth Analysis

Interactive Excel Dashboard Edition | Companion workbook: HR_Workforce_Analytics_Dashboard.xlsx

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

In today's competitive business environment, organizations increasingly rely on data-driven decision-making to improve workforce productivity, employee retention, and overall organizational performance. Human Resource Analytics, also known as People Analytics, has emerged as a critical discipline that uses data analysis techniques to understand workforce trends and guide strategic HR decisions.

Employee attrition remains one of the most pressing challenges faced by organizations globally. High turnover can result in increased recruitment costs, reduced productivity, loss of institutional knowledge, and lower employee morale. Similarly, job satisfaction, salary structure, career progression, and departmental performance all play essential roles in shaping workforce stability.

This project applies HR analytics techniques to a workforce dataset of 800 employees to uncover patterns related to attrition, job satisfaction, salary, departmental performance, and promotion opportunities. The entire solution is built in Microsoft Excel as a live, formula-driven dashboard: every figure, chart and written insight recalculates automatically when the underlying data or the dashboard filters change.

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1.2 Problem Statement

Many organizations struggle to identify the root causes of employee turnover and dissatisfaction. Traditional HR reporting often relies on static reports and manually refreshed pivot tables, which quickly go out of date and encourage decisions based on assumptions rather than evidence. Without timely analysis, companies may fail to recognize departments with elevated attrition, pay imbalances, or career stagnation.

This project addresses these challenges by analyzing workforce data to identify the factors associated with attrition, comparing compensation and promotion outcomes across groups, assessing departmental performance, and delivering a live monitoring dashboard that HR managers can filter and refresh without technical effort.

1.3 Aim of the Study

The aim of this project is to analyze workforce data to understand employee behavior, identify the factors associated with attrition, and provide data-driven recommendations that improve organizational performance and employee retention, delivered through a professional, interactive Excel dashboard.

1.4 Objectives of the Study

- To evaluate overall workforce size, structure and distribution across departments.

- To analyze employee attrition and identify where and among whom turnover is highest.
- To assess the relationship between salary, job satisfaction and retention.
- To conduct department-wise analysis to identify performance gaps.
- To evaluate promotion opportunities and career growth patterns.
- To build a live, filter-driven dashboard that updates automatically when data changes, with built-in data-quality checks.

1.5 Significance of the Study

This study demonstrates how organizations can use workforce data to make strategic HR decisions. It provides insights that can help reduce employee turnover, improve satisfaction, refine compensation policies, and strengthen career development frameworks. Because the dashboard is fully formula-driven, it can be reused each reporting cycle by simply updating the data.

Additionally, the project gives aspiring data analysts' practical experience in solving real-world business problems with an HR dataset, from data validation through to executive-style dashboard design.

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CHAPTER TWO: DATA DESCRIPTION & METHODOLOGY

2.1 Dataset Description

The dataset is an HR Workforce Analytics dataset containing 800 employee records across five departments: Finance (170 employees), HR (174), IT (159), Operations (142) and Sales (155). Each record holds eight fields, summarized in Table 2.1.

Table 2.1: Variables in the dataset

Variable	Description	Values in the data
Employee ID	Unique employee identifier	EMP1000 – EMP1799 (800 unique IDs)
Age	Employee age in years	21 – 59
Department	Employee's department	Finance, HR, IT, Operations, Sales
Salary	Annual salary (USD)	$30,029 – $119,971
Years At Company	Tenure in whole years	0 – 19
Job Satisfaction	Satisfaction rating	1 (lowest) – 4 (highest)
Promotion Last 5Years	Promoted in the last 5 years	0 = No, 1 = Yes
Attrition	Whether the employee left	Yes (146) / No (654)

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Gender and education level are not recorded in this dataset, so the analysis focuses on the eight variables above.

2.2 Data Preparation

The dataset was loaded into Microsoft Excel as a structured Excel Table (tblHR) and validated before analysis. The following steps were taken:

- **Duplicate check:** confirmed that all 800 Employee IDs are unique (duplicates are also highlighted automatically in red on the Data sheet).
- **Missing-value check:** confirmed that none of the eight required fields contains blanks.
- **Value validation:** confirmed that Attrition contains only Yes/No, Job Satisfaction only 1–4, and Promotion only 0/1; drop-down validation now enforces this for new entries.
- **Standardization:** department names are drawn from a single master list on the Settings sheet.
- **Clean up:** removed the old helper columns (Total Employees, Employees Left, Attrition Rate, and Column1) from the data sheet, because these values are now calculated automatically.
- **Banding:** tenure, age and salary were grouped into editable bands (for example 0–2, 3–5, 6–10, 11–15 and 16+ years) so patterns are readable.

2.3 Methodology and Workbook Design

Rather than relying on pivot tables, which must be refreshed manually, the analysis uses Excel formulas (COUNTIFS, AVERAGEIFS and INDEX/MATCH) that read directly from the Data sheet. The workbook is organized into five sheets, described in Table 2.2.

Table 2.2: Structure of the Excel workbook

Sheet	Purpose
Dashboard	Executive view with three dropdown filters, six KPI cards, auto-written insights, 14 charts, a heatmap and a department scorecard.
Data	The 800-record dataset as an Excel Table, with entry validation and duplicate highlighting.
Settings	Department list, satisfaction scale, tenure / age / salary band limits, the attrition target and automatic data-quality checks.
Calc	The calculation engine that feeds every KPI and chart from the Data sheet.
Guide	Instructions for updating and maintaining the workbook.

Key metric definitions are as follows. Attrition rate is the number of employees who left divided by headcount for the selected group. Promotion rate is the share of employees promoted in the last five years. Average satisfaction is the mean of the 1–4 rating. An attrition target of 15% is used as a benchmark; this is a placeholder assumption that can be changed on the Settings sheet to match an organization's approved target.

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CHAPTER THREE: ANALYSIS & FINDINGS

3.1 Workforce Overview

The organization employs 800 people. Of these, 146 have left, giving an overall attrition rate of 18.3%, which is above the 15% benchmark. The average salary is $75,062, the average job satisfaction rating is 2.50 out of 4, the average tenure is 9.8 years, and 261 employees (32.6%) were promoted in the last five years.

The workforce is well balanced across departments, with headcounts ranging from 142 (Operations) to 174 (HR). Table 3.1 summarizes the main indicators by department.

Table 3.1: Department scorecard

Department	Employees	Left	Attrition	Avg salary	Avg satis.	Promoted	Tenure
Finance	170	39	22.9%	$74,193	2.54	34.7%	8.7
HR	174	31	17.8%	$73,784	2.55	30.5%	9.5
IT	159	21	13.2%	$77,439	2.48	36.5%	10.2
Operations	142	23	16.2%	$75,429	2.39	31.0%	10.1
Sales	155	32	20.6%	$74,673	2.50	30.3%	10.5
All departments	**800**	**146**	**18.3%**	**$75,062**	**2.50**	**32.6%**	**9.8**

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3.2 Attrition Analysis

By department. Finance has the highest attrition (22.9%), followed by Sales (20.6%), HR (17.8%) and Operations (16.2%). IT is the only department below the 15% benchmark, at 13.2%.

By job satisfaction. Employees who rated their satisfaction 1 left at 19.4%, compared with 16.6% for those who rated it 4. The relationship exists but is modest: a gap of 2.8 percentage points and the highest rate (19.9%) actually occurs at rating 3. Satisfaction therefore appears to be a contributing factor rather than a strong, standalone driver of attrition.

Table 3.2: Attrition by job satisfaction rating

Rating	Employees	Left	Attrition rate
1 (lowest)	216	42	19.4%
2	182	31	17.0%
3	191	38	19.9%
4 (highest)	211	35	16.6%

By tenure. Attrition is highest among employees with 3–5 years of service (20.9%) and stays near 20% through 15 years, before dropping to 13.4% for those with 16 or more years. Long-serving employees are the most stable group, while the mid-career years are the main risk period.

Table 3.3: Attrition by tenure band

Tenure	Employees	Left	Attrition rate
0–2 years	121	21	17.4%
3–5 years	110	23	20.9%
6–10 years	186	37	19.9%
11–15 years	211	42	19.9%
16+ years	172	23	13.4%

By age. Employees aged 40–49 show the highest attrition (21.2%), while those aged 50 and over show the lowest (15.9%). Under-30s (17.6%) and 30–39s (18.2%) sit in between.

3.3 Salary Impact Analysis

Salary does not explain attrition on its own. Employees who left earned an average of $75,981, slightly more than the $74,857 earned by those who stayed (about 1.5% higher). Attrition by pay band follows a U-shape rather than a straight line: the lowest-paid group (under $50k) leaves at 20.7%, the middle bands ($50k–$90k) are lowest at 15.3%, and the highest-paid group ($110k+) has the highest rate of all at 24.3%.

Table 3.4: Attrition by salary band

Salary band	Employees	Left	Attrition rate
Under $50k	193	40	20.7%
$50k – $70k	177	27	15.3%
$70k – $90k	150	23	15.3%
$90k – $110k	165	28	17.0%
$110k+	115	28	24.3%

Average salary by department is highest in IT ($77,439) and lowest in HR ($73,784). Average salary by satisfaction rating ranges from $71,878 (rating 3) to $77,714 (rating 2) with no consistent upward pattern, which suggests that non-monetary factors such as career growth, management and workload matter alongside pay. The high attrition among top earners deserves further investigation, as it may reflect external job-market pull rather than dissatisfaction with pay.

3.4 Department-Wise Analysis

Departments differ noticeably. Finance combines the highest attrition (22.9%) with the lowest average tenure (8.7 years), suggesting a churn problem concentrated among newer staff. Sales is the second highest in attrition (20.6%) and has the lowest promotion rate (30.3%). IT is the strongest performer, with the lowest attrition (13.2%), the highest average salary and the highest promotion rate (36.5%). Operations records the lowest average satisfaction (2.39).

The department-by-satisfaction heat map on the dashboard highlights specific hot spots, for example Finance employees rating satisfaction 2 (32.3%) or 1 (29.2%), IT employees rating 3 (29.4%) and Sales employees rating 4 (26.3%). Each cell contains roughly 30–50 employees, so these figures should be treated as indicators for follow-up rather than firm conclusions.

3.5 Promotion & Career Growth Analysis

Only 261 of 800 employees (32.6%) received a promotion in the last five years, meaning two out of three had no progression. Promotion is associated with better retention: promoted employees left at 16.1% (42 of 261) compared with 19.3% (104 of 539) for those not promoted, a difference of 3.2 percentage points.

Promotion rates vary by department, from 36.5% in IT to 30.3% in Sales, and by tenure. Employees with 3–5 years of service have the lowest promotion rate (23.6%), which coincides with their elevated attrition (20.9%) and points to a possible career-progression gap at this stage. Promotion rates are highest for the 11–15 year group (37.4%). Interestingly, promoted employees earn slightly less on average ($73,730) than non-promoted employees ($75,707), indicating that promotions are not translating into a clear pay premium in this dataset.

3.6 Automated Insights and Data Quality

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To make the findings repeatable, the dashboard generates five plain-language insight sentences automatically from the current selection, for example the attrition rate against the target, the highest- and lowest-attrition departments, the satisfaction gap, the salary comparison between leavers and stayers, and the promotion comparison. These sentences rewrite themselves whenever the data or filters change.

A data-quality panel on the Settings sheet checks for departments missing from the master list, invalid attrition, satisfaction or promotion values, and missing entries. If any issue is detected, the dashboard banner changes from “data checks passed” to a warning, so users know the figures need review before they are shared.

CHAPTER FOUR: DASHBOARD DEVELOPMENT

4.1 Dashboard Components

The HR dashboard is a single, scrollable Excel page with a consistent navy-and-white design. It contains the following components:

- **Filter bar:** three dropdowns (Department, Job Satisfaction, Promotion) that instantly update every number and chart.
- **Employee Count KPI:** total headcount for the current selection, with its share of the workforce.
- **Employees Left and Attrition Rate KPIs:** the attrition rate turns red when it is above the target and green when within it.
- **Average Salary KPI:** with a comparison against the company average or against leavers.
- **Job Satisfaction KPI:** average rating on the 1–4 scale.
- **Promotion Rate KPI:** share of employees promoted in the last five years.
- **Key Insights panel:** five automatically written sentences that describe the current selection.

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4.2 Visualizations

The dashboard presents 14 charts, a heat map and a scorecard in three themed sections, below a header with the filters, KPI cards and key insights. Figures 4.1 to 4.4 on the following pages show each part of the dashboard exactly as it appears in Excel.

A. Dashboard Header, KPI Cards and Key Insights

The top of the dashboard holds the live status banner, the three filters, the six KPI cards and the five auto-written insights.

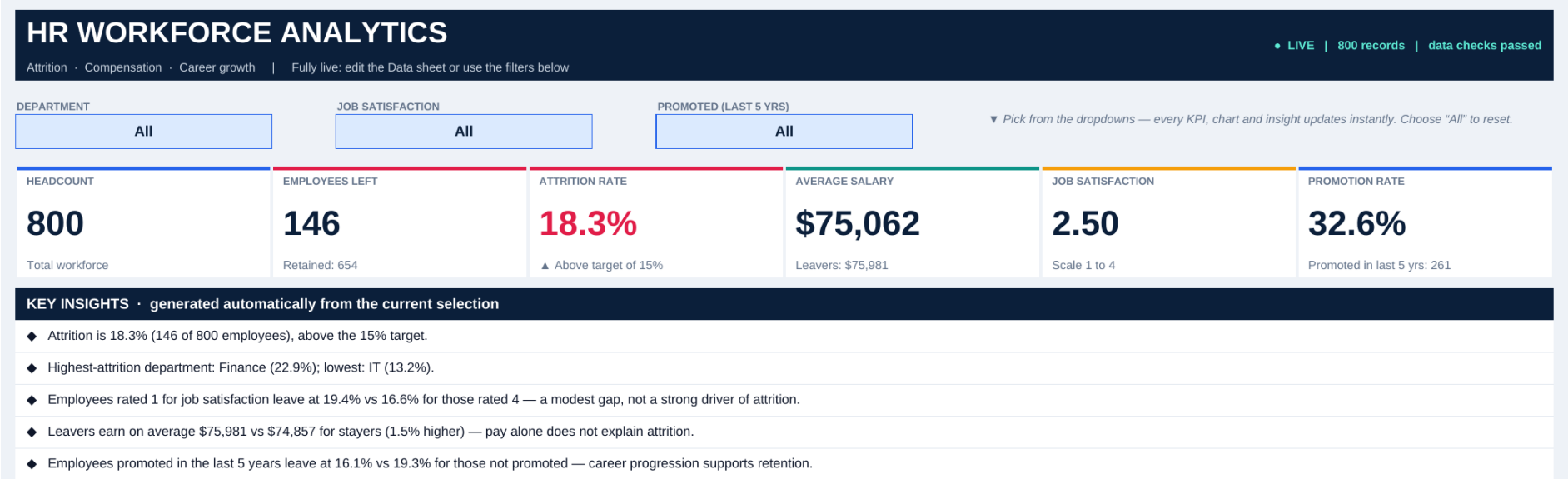


Figure 4.1: Dashboard header, filters, KPI cards and key insights

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B. Employee Attrition Analysis

This section shows attrition rate by department (ranked), leavers by job satisfaction rating, attrition by tenure and by age band, the stayed-versus-left retention split, and the department-by-satisfaction attrition heat map.

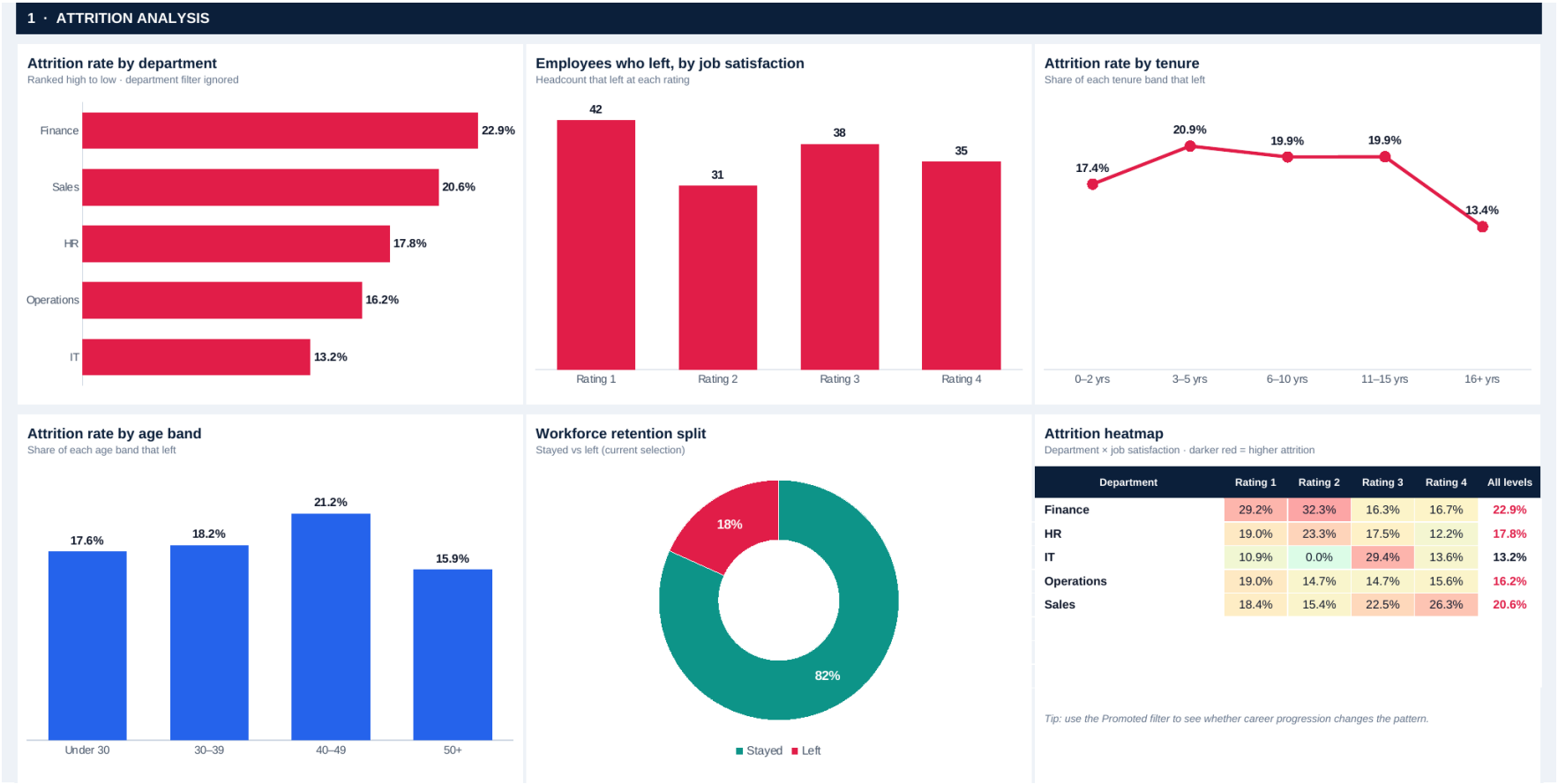


Figure 4.2: Employee Attrition Analysis section

C. Compensation & Attrition (Salary Impact on Employee Behavior)

This section compares average salary for stayers and leavers, average salary by department, satisfaction rating, tenure and age band, and attrition rate by salary band.



Figure 4.3: Compensation & Attrition section

D. Promotion & Career Growth, and Department Scorecard

This section compares attrition for promoted and non-promoted employees, promotion rate by department and by tenure, and closes with the department scorecard, which highlights the department selected in the filter bar.

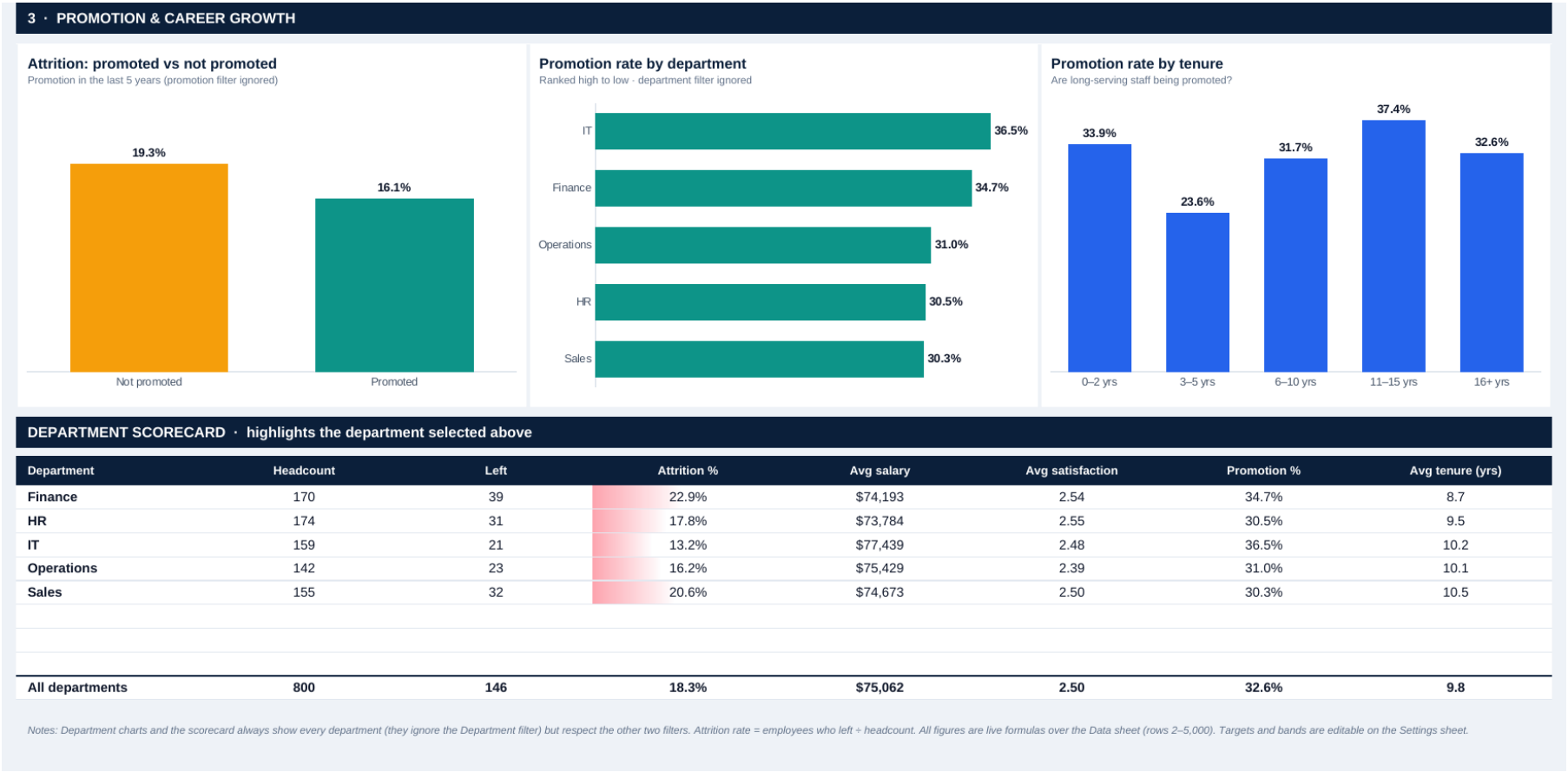


Figure 4.4: Promotion & Career Growth section and Department Scorecard

4.3 How the Dashboard Stays Live

- **Formula-driven:** all KPIs and charts read from the Data sheet through COUNTIFS and AVERAGEIFS formulas, so editing a value or adding a row updates everything (the dashboard reads up to 5,000 rows).
- **Interactive filters:** each chart ignores the filter for its own dimension, so comparisons stay visible while the rest of the page responds to the selection.
- **Self-checking:** automatic data-quality checks and a status banner protect the integrity of the figures.
- **Configurable:** the department list, band limits and attrition target are editable on the Settings sheet.

Adding a new department requires typing it into the Settings list; the KPIs, heat map and scorecard pick it up automatically (up to eight departments), while the three ranked department charts need their data range extended by one row.

4.4 Dashboard Value

The dashboard gives management live workforce insights for proactive decision-making. Because it replaces static, manually refreshed pivot tables with a formula-driven design, HR teams can update the data, apply filters and read the refreshed findings in seconds, with no technical expertise required.

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CHAPTER FIVE: CONCLUSION & RECOMMENDATIONS

5.1 Conclusion

This project demonstrates the value of HR analytics in understanding workforce behavior and improving organizational decision-making. Overall attrition of 18.3% is above the 15% benchmark, and four of the five departments exceed it. Attrition is concentrated in Finance and Sales, among employees in the mid-career years (3–15 years of service) and aged 40–49, and among both the lowest-paid and the highest-paid groups.

The evidence shows that attrition is not driven by a single factor. Lower satisfaction and the absence of a recent promotion are each associated with higher attrition, but the effects are moderate, and salary level alone does not explain who leaves. This point to a combination of career growth, engagement and role-specific factors that warrant targeted action rather than a blanket pay response.

The Excel dashboard turns these findings into a reusable tool: it updates automatically, can be filtered by department, satisfaction and promotion status, writes its own summary insights, and monitors data quality.

5.2 Recommendations

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- **Prioritize Finance and Sales.** Conduct exit and stay interviews, review workload and leadership practices, and set department-level attrition targets.
- **Strengthen mid-career retention.** Introduce development plans, mentoring and clear advancement paths for employees with 3–15 years of service, where attrition is highest.
- **Make promotion policies transparent.** Only a third of employees were promoted in five years; clear criteria and timelines should be established, especially for the 3–5 year group.
- **Review pay competitiveness selectively.** Benchmark roles in the under-$50k band and investigate why the $110k+ group has the highest attrition, rather than raising pay across the board.
- **Act on satisfaction feedback.** Follow up low-rating groups, starting with the heat map hot spots, through engagement programs and manager coaching.
- **Adopt the dashboard as a monthly routine.** Update the Data sheet each cycle, confirm the data-quality banner, and review attrition against an approved target set on the Settings sheet.

5.3 Limitations and Further Work

The dataset is a single snapshot, so the findings describe associations and not proven causes. Some heat map cells contain small numbers of employees, and the dataset does not include gender, education or exit reasons. Future work could extend the project with a predictive risk

model, for example logistic regression, to score individual employees' likelihood of leaving, and could add time-series data to track attrition trends month by month.

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PROJECT IMPACT

This project positions the analyst as a business-focused data professional capable of solving real organizational problems using workforce data. It demonstrates skills in:

- Data cleaning and validation
- Advanced Excel formulas and structured data modeling
- Business intelligence and interactive dashboard development
- Data storytelling and evidence-based recommendations

Strategic Recommendation

Organizations should treat workforce analytics as an ongoing process rather than a one-off report. Maintaining a live dashboard, reviewing it regularly against an agreed attrition target, and acting, early on the departments and career stages where attrition is concentrated will allow HR leaders to reduce turnover, protect institutional knowledge and build a more stable, engaged workforce.

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